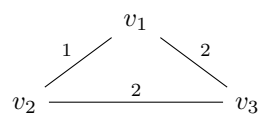
Research Statement

Shuai Wei

1 Introduction

My interests lie in commutative algebra, with emphasis on its connections to combinatorics. I mainly work with monomial ideals in the polynomial ring $R = \mathbb{K}[X_1, \dots, X_d]$ with $\mathbb{K}$ a field, specifically, *edge ideals* of *edge-weighted* graphs G_ω where $G = (V, E)$ is a finite simple graph with $V = \{v_1, \dots, v_d\}$ and a weight function $\omega : E \to \mathbb{N}$ with $\mathbb{N} = \{1, 2, \dots\}$, as in the following example.

Example 1.1. Let G_ω be the following weighted graph with $V(G) = \{v_1, v_2, v_3\}$ and $E(G) = \{v_1v_2, v_2v_3, v_1v_3\}$ and $\omega(v_1v_2) = 1$, $\omega(v_2v_3) = 2$ and $\omega(v_1v_3) = 2$.



Then the weighted edge ideal [8] of G_ω is given by

$$I(G_\omega) = (X_1X_2, X_2^2X_3^2, X_1^2X_3^2)R.$$

To study such ideals, in my dissertation I introduce the notion of "generalized $\mathbb{N}$-weighted simplicial complex". This is the topic of Section 2 below where I construct a weighted version of Stanley-Reisner correspondence, and if it is "bounded above", I give an irredundant decompositon of the weighted Stanley-Reisner ideal in terms of the defined simplicial complex. At the end of this section, I define the Alexander dual in this setting and see how it is related to the dual of a monomial ideal, defined by E. Miller [6].

In Section 3, I discusss how I compute the Cohen-Macaulay type of weighted edge ideals combinatorially for certain edge-weighted graphs.

2 *Generalized $\mathbb{N}$-weighted simplicial complex*

The Stanley-Reisner correspondence uses simplicial complexes to study square-free monomial ideals. In order to use similar techniques to study certain non-square-free monomial ideals, in my dissertation, I define a weighted version which I call a *generalized $\mathbb{N}$-weighted simplicial complex*.

A *generalized $\mathbb{N}$-weighted simplicial complex* on V is a non-empty collection Υ of sets of form (V', δ') with $V' \subseteq V$ and $\delta' : V' \to \mathbb{N}$ that is closed under $\preccurlyeq$: For subsets $V_1', V_2' \subseteq V'$ and functions $\delta_1' : V_1' \to \mathbb{N}$ and $\delta_2' : V_2' \to \mathbb{N}$, we write $(V_1', \delta_1') \preccurlyeq (V_2', \delta_2')$ if $V_1' \subseteq V_2'$ and $\delta_1' \leq \delta_2'|_{V_2'}$.

For a subset $V' \subseteq V$ and a function $\delta' : V' \to \mathbb{N}$, we notationally set

$$(V', \delta') = \{v_i^{\delta'(v_i)} \mid v_i \in V'\},$$

where $\delta'(v_i)$ is the *weight* of v_i.

For instance,

$$\Upsilon = \{\{v_2^2, v_3\}, \{v_2, v_3^2\}, \{v_2, v_3\}, \{v_2^2\}, \{v_3^2\}, \{v_1\}, \{v_2\}, \{v_3\}, \emptyset\}$$

is a generalized $\mathbb{N}$-weighted simplicial complex.

Associated to Υ, we define *weighted Stanley-Reisner ideal* J_Υ by

$$\begin{aligned} J_\Upsilon &= \big(\underline{X}^{(V',\delta')} \mid V' \subseteq V,\ \delta' : V' \to \mathbb{N} \text{ and } (V',\delta') \notin \Upsilon\big)R \\ &= \left(\prod_{v_i \in V'} X_i^{\delta'(v_i)} \mid V' \subseteq V,\ \delta' : V' \to \mathbb{N} \text{ and } (V',\delta') \notin \Upsilon\right) R. \end{aligned}$$

If $\delta' : V' \to \mathbb{N}$ is the constant function 1 for all $(V',\delta') \in \Upsilon$, then Υ becomes a usual simplicial complex Δ, and this recovers the definition of the original notion of a Stanley-Reisner ideal.

For example, given the generalized $\mathbb{N}$-weighted simplicial complex

$$\Upsilon = \{\{v_2^2, v_3\}, \{v_2, v_3^2\}, \{v_2, v_3\}, \{v_2^2\}, \{v_3^2\}, \{v_1\}, \{v_2\}, \{v_3\}, \emptyset\},$$

we have

$$J_\Upsilon = (X_2^2 X_3^2, X_1X_2, X_1X_3)R.$$

Let $I \subseteq R$ be a monomial ideal [7], for $j = 1, \ldots, d$ assume M_j is the maximal exponent of X_j on the minimal generators of I. We define

$$\Upsilon(I) = \{(V',\delta') \text{ with } V' \subseteq V,\ \delta' : V' \to \mathbb{N} \text{ and } \delta'(v_i) \leq M_i\ \forall\ v_i \in V' \mid \underline{X}^{(V',\delta')} \notin I\},$$

which is then a generalized $\mathbb{N}$-weighted simplicial complex. The next result from my dissertation is a weighted version of the Stanley-Reisner correspondence.

Theorem 2.1. *(a) Let $I \subseteq R$ be a monomial ideal [7], for $j = 1, \ldots, d$ assume M_j is the maximal exponent of X_j on the minimal generators of I. Then*

$$J_{\Upsilon(I)} = I.$$

(b) Let Υ be a generalized $\mathbb{N}$-weighted simplicial complex such that for $i = 1, \ldots, d$ we have $M_i^\Upsilon := \max\{\delta'(v_i) \mid v_i \in V'$ and $(V',\delta') \in \Upsilon\} < \infty$. Then

$$\Upsilon(J_\Upsilon) = \Upsilon.$$

For instance, with $J_\Upsilon = (X_2^2X_3^2, X_1X_2, X_1X_3)R$, we have

$$\begin{aligned} \Upsilon(J_\Upsilon) &= \{(V',\delta') \text{ with } V' \subseteq V \text{ and } \delta' : V' \to \mathbb{N} \mid \underline{X}^{(V',\delta')} \notin J_\Upsilon\} \\ &= \{\{v_2^2, v_3\}, \{v_2, v_3^2\}, \{v_2, v_3\}, \{v_2^2\}, \{v_3^2\}, \{v_1\}, \{v_2\}, \{v_3\}, \emptyset\} \\ &= \Upsilon. \end{aligned}$$

In my dissertation, I show that, as in the classical setting, these yield an irrredundant irreducible decomposition of $J_{\Upsilon^{\mathbf{M}}}$:

Theorem 2.2. *Let Υ be a generalized $\mathbb{N}$-weighted simplicial complex such that for $i = 1, \ldots, d$ we have $M_i^{\Upsilon} := \max\{\delta'(v_i) \mid v_i \in V' \text{ and } (V', \delta') \in \Upsilon\} < \infty$. Then the ideal J_{Υ} has the following irreducible (primary) decomposition*

$$J_{\Upsilon} = \bigcap_{face\ \rho \in \Upsilon} Q_{\rho} = \bigcap_{\rho\ facet} Q_{\rho},$$

where $Q_{\rho} = (X^{\tau_{\rho}(v_i)} \mid v_i \in (V \smallsetminus V') \sqcup W_{\rho})R$, $W_{\rho} = \{v_i \in V' \mid \delta'(v_i) < M_i^{\Upsilon}\}$ and $\tau_{\rho} : (V \smallsetminus V') \sqcup W_{\rho} \to \mathbb{N}$ is given by

$$\tau_{\rho}(v_i) = \begin{cases} 1 & \text{if } v_i \in V \smallsetminus V', \\ \delta'(v_i) + 1 & \text{if } v_i \in W_{\rho}. \end{cases}$$

The second intersection is irredundant.

For instance, with

$$\Upsilon = \{\{v_2^2, v_3\}, \{v_2, v_3^2\}, \{v_2, v_3\}, \{v_2^2\}, \{v_3^2\}, \{v_1\}, \{v_2\}, \{v_3\}, \emptyset\},$$

and $M_1^{\Upsilon} = 1$, $M_2^{\Upsilon} = 2$ and $M_3^{\Upsilon} = 2$, we have

$$\begin{aligned} J_{\Upsilon} &= \bigcap_{\text{face } \rho \in \Upsilon} Q_{\rho} \\ &= Q_{\{v_2^2, v_3\}} \cap Q_{\{v_2, v_3^2\}} \cap Q_{\{v_2, v_3\}} \cap Q_{\{v_2^2\}} \cap Q_{\{v_3^2\}} \cap Q_{\{v_1\}} \cap Q_{\{v_2\}} \cap Q_{\{v_3\}} \cap Q_{\emptyset} \\ &= (X_1, X_3^2)R \cap (X_1, X_2^2)R \cap (X_1, X_2^2, X_3^2)R \cap (X_1, X_3)R \cap (X_1, X_2)R \cap (X_2, X_3)R \\ &\quad \cap (X_1, X_3)R \cap (X_1, X_2)R \cap (X_1, X_2, X_3)R \\ &= (X_1, X_3^2)R \cap (X_1, X_2^2)R \cap (X_2, X_3)R \\ &= Q_{\{v_2^2, v_3\}} \cap Q_{\{v_2, v_3^2\}} \cap Q_{\{v_1\}} \\ &= \bigcap_{\text{facet } \rho \in \Upsilon} Q_{\rho}. \end{aligned}$$

Much of our interest in this area focuses on the edge ideal $I(G)$ of a graph G [4] in some weighted settings. For the classical construction, this is the same as the Stanley-Reisner ideal J_{Δ_G} of the independence complex Δ_G; this is the set of all independent subsets of V where $W \subseteq V$ is *independent* if no two vertices in W are adjacent in G. In my dissertation I introduce and investigate the following weighted version of this notion.

Let G_{ω} be an edge weighted graph on V. We say that an ordered pair (V', δ') with $V' \subseteq V$ and $\delta' : V' \to \mathbb{N}$ satisfying $\delta'(v_i) \leq \max\{\omega(v_i v_j) \mid v_i v_j \in E\}$ for any $v_i \in V'$ is a *weighted independent* set in G_{ω} if for any $v_i, v_j \in V'$, either $v_i v_j \notin E$, or $\min\{\delta'(v_i), \delta'(v_j)\} < \omega(v_i v_j)$. Let Υ_{ω} denote the set of weighted independent sets (V', δ') in G_{ω}. This is the *weighted independence complex* of G_{ω}.

The weighted independent sets in G_{ω} from Example 1.1 are by definition

$$\Upsilon_{\omega} = \{\{v_1^2, v_3\}, \{v_1, v_3^2\}, \{v_1, v_3\}, \{v_2^2, v_3\}, \{v_2, v_3^2\}, \{v_2, v_3\}, \{v_1^2\}, \{v_2^2\}, \{v_3^2\}, \{v_1\}, \{v_2\}, \{v_3\}, \emptyset\}.$$

The weighted edge ideal $I(G_{\omega})$ can be decomposed in terms of weighted vertex covers of G_{ω} by [8]. In my dissertation I give the following decomposition result for these ideals in terms of weighted independent sets.

Theorem 2.3. *We have $M_i^{\Upsilon_\omega} = \max\{\omega(v_iv_j) \mid v_iv_j \in E\}$ for $i = 1, \dots, d$. The ideal J_{Υ_ω} has the following irreducible (primary) decompositions*

$$I(G_\omega) = J_{\Upsilon_\omega} = \bigcap_{\rho \in \Upsilon_\omega} Q_\rho = \bigcap_{maximal\ \rho \in \Upsilon_\omega} Q_\rho.$$

The second intersection is irredundant.

For instance, in Example 1.1, with $M_1^{\Upsilon_\omega} = 2$, $M_2^{\Upsilon_\omega} = 2$ and $M_3^{\Upsilon_\omega} = 2$, we have

$$\begin{aligned}
J_{\Upsilon_\omega} &= \bigcap_{\text{face } \rho \in \Upsilon_\omega} Q_\rho \\
&= Q_{\{v_1^2, v_3\}} \cap Q_{\{v_1, v_3^2\}} \cap Q_{\{v_1, v_3\}} \cap Q_{\{v_2^2, v_3\}} \cap Q_{\{v_2, v_3^2\}} \cap Q_{\{v_2, v_3\}} \\
&\cap Q_{\{v_1^2\}} \cap Q_{\{v_2^2\}} \cap Q_{\{v_3^2\}} \cap Q_{\{v_1\}} \cap Q_{\{v_2\}} \cap Q_{\{v_3\}} \cap Q_\emptyset \\
&= (X_2, X_3^2)R \cap (X_1^2, X_2)R \cap (X_1^2, X_2, X_3^2)R \cap (X_1, X_3^2)R \cap (X_1, X_2^2)R \cap (X_1, X_2^2, X_3^2)R \\
&\cap (X_2, X_3)R \cap (X_1, X_3)R \cap (X_1, X_2)R \cap (X_2, X_3)R \cap (X_1, X_3)R \cap (X_1, X_2)R \cap (X_1, X_2, X_3)R \\
&= (X_2, X_3^2) \cap (X_1^2, X_2) \cap (X_1, X_3^2)R \cap (X_1, X_2^2)R \\
&= Q_{\{v_1^2, v_3\}} \cap Q_{\{v_1, v_3^2\}} \cap Q_{\{v_2^2, v_3\}} \cap Q_{\{v_2, v_3^2\}} \\
&= \bigcap_{\text{facet } \rho \in \Upsilon_\omega} Q_\rho.
\end{aligned}$$

Let Υ be a generalized $\mathbb{N}$-weighted simplicial complex such that for $i = 1, \dots, d$ we have $M_i^{\Upsilon} < \infty$. Define the *Alexander dual* $\Upsilon^\vee$ of Υ by

$$\Upsilon^\vee = \{\rho := (V', \delta') \text{ with } V' \subseteq V,\ \delta' : V' \to \mathbb{N} \mid ((V \smallsetminus V') \sqcup W_\rho, \varphi_\rho) \notin \Upsilon\},$$

where $\varphi_\rho : (V \smallsetminus V') \sqcup W_\rho \to \mathbb{N}$ is defined by

$$\varphi_\rho(v_i) = \begin{cases} M_i & \text{if } v_i \in V \smallsetminus V', \\ M_i - \delta'(v_i) & \text{if } v_i \in W_\rho. \end{cases}$$

Then $\Upsilon^\vee$ is a generalied $\mathbb{N}$-weighted simplicial complex.

For instance, the dual of the weighted independence complex Υ_ω for Example 1.1 is

$$\Upsilon_\omega^\vee = \{\{v_1, v_3^2\}, \{v_2, v_3^2\}, \{v_1, v_3\}, \{v_1, v_2\}, \{v_2, v_3\}, \{v_1\}, \{v_2\}, \{v_3^2\}, \{v_3\}, \emptyset\}.$$

Then

$$J_{\Upsilon_\omega^\vee} = (X_1^2X_2, X_1X_2^2, X_1^2X_3, X_2^2X_3)R.$$

In addition, I introduce and investigate a variation of E. Miller's algebraic Alexander duality [6] in this context which is slightly better behaved in some respects.

Let $I \subseteq R$ be a monomial ideal, for $j = 1, \dots, d$ assume M_j and m_j are the maximal and minimal exponent of X_j on the minimal generators of I, respectively. Write $I = \bigcap_{i=1}^n P(V_i', \delta_i')$ as a irredundant irreducible decomposition, where $P(V_i', \delta_i') = (X_{j_i}^{\delta_i'(v_{j_i})} \mid v_{j_i} \in V_i')R$. Define the *Alexander dual* $I^\vee$ of I by

$$I^\vee = (\underline{X}^{(V_i', \delta_i'')} \mid i = 1, \dots, n),$$

where for $i = 1, \ldots, n$: $\delta_i''(v_j) = M_j + m_j - \delta_i'(v_j)$ for any $v_j \in V_i'$ and $\underline{X}^{(V_i', \delta_i'')} = \prod_{v_i \in V_i'} X_i^{\delta''(v_i)}$. If all the m_j's are 1 in the above definition, then it recovers the classical Alexander dual. In terms of this definition, I am working on proving that $I^{\vee\vee} = I$.

In some situations, E. Miller's definition works better. So for the rest of this section I'll use his duality definition for monomial ideals. I justify the name of the previous construction by showing in the next two results from my dissertation that Alexander duality commutes with the weighted Stanley-Reisner correspondence.

Theorem 2.4. *Let Υ be a generalized $\mathbb{N}$-weighted simplicial complex such that for $i = 1, \ldots, d$ we have $M_i^{\Upsilon} < \infty$. Then*

$$J_{\Upsilon}^{\vee} = J_{\Upsilon^{\vee}}.$$

For instance, consider the weighted independence complex Υ_ω for Example 1.1. Since the irredundant irreducible decomposition of $I(G_\omega)$ is

$$I(G_\omega) = (X_1X_2, X_2^2X_3^2, X_1^2X_3^2) = (X_1, X_2^2)R \cap (X_1^2, X_2)R \cap (X_1, X_3^2)R \cap (X_2, X_3^2)R,$$

we have

$$J_{\Upsilon_\omega}^{\vee} = I(G_\omega)^{\vee} = (X_1^2X_2, X_1X_2^2, X_1^2X_3, X_2^2X_3)R = J_{\Upsilon_\omega^{\vee}}.$$

Theorem 2.5. *If $I \subseteq R$ is a monomial ideal, then*

$$\Upsilon(I^{\vee}) = \Upsilon(I)^{\vee}.$$

For instance, consider the weighted edge ideal $I(G_\omega)$ for Example 1.1. We have by Macaulay2

$$I(G_\omega)^{\vee} = (X_1^2X_2, X_1X_2^2, X_1^2X_3, X_2^2X_3).$$

Then

$$\Upsilon(I(G_\omega)^{\vee}) = \{\{v_1, v_2\}, \{v_1, v_3^2\}, \{v_1, v_3\}, \{v_2, v_3^2\}, \{v_2, v_3\}, \{v_3^2\}, \{v_1\}, \{v_2\}, \{v_3\}, \emptyset\}.$$

We can show that $\Upsilon(I(G_\omega)) = \Upsilon_\omega$ generally. In this particular example, we have

$$\begin{aligned}\Upsilon(I(G_\omega))^{\vee} &= \Upsilon_\omega^{\vee} \\ &= \{\{v_1, v_3^2\}, \{v_2, v_3^2\}, \{v_1, v_3\}, \{v_1, v_2\}, \{v_2, v_3\}, \{v_1\}, \{v_2\}, \{v_3^2\}, \{v_3\}, \emptyset\} \\ &= \Upsilon(I(G_\omega)^{\vee}).\end{aligned}$$

3 *Cohen-Macaulay type of weighted edge ideals*

The *type* [2] of a graded Cohen-Macaulay quotient ring R/I

$$\mathrm{r}_R(R/I) = \dim_{\mathbb{K}}(\mathrm{Ext}_R^n(\mathbb{K}, R/I)) \text{ [9]}$$

quantifies important aspects of the minimal injective resolution of R/I and is accordingly a measure of how close the ring is to being Gorenstein [1]. However, this invariant is generally quite difficult to compute. In my dissertation, I explicitly calculate this number combinatorially when I is a monomial ideal constructed from an edge-weighted graph.

For an edge-weighted graph G_ω, Paulsen and Sather-Wagstaff introduce a notion of weighted vertex cover and minimal weighted vertex cover [8] that is similar to the unweighted notions, but that also tracks weighted information. For instance, the minimal weighted vertex covers for the weighted 2-path

$$(P_2)_\epsilon = (\ v_1 \overset{2}{\text{———}} v_2 \overset{3}{\text{———}} v_3)$$

are indicated graphically by circling the vertices in the cover and giving them weights that "cover" all the weighted edges of the graph:

$$v_1^2 \overset{2}{\text{———}} v_2 \overset{3}{\text{———}} v_3^3 \qquad v_1 \overset{2}{\text{———}} v_2^2 \overset{3}{\text{———}} v_3.$$

Paulsen and Sather-Wagstaff [8] also characterize the weighted trees that are Cohen-Macaulay. Their characterization is similar to Villarreal's description of the unweighted Cohen-Macaulay trees [10]. In particular, they are all weighted "suspensions," which we write as $G_\omega = (\Sigma H)_\omega$ where H_ϵ is an induced weighted subgraph. In the following theorem in my dissertation, I show how to compute the type of $R'/I((\Sigma H)_\omega)$ combinatorially in terms of the information of H_ϵ.

Theorem 3.1. *Let $(\Sigma H)_\omega$ be a Cohen-Macaulay weighted suspension [8] of H_ϵ such that $\omega(v_iv_j) \leq \omega(v_iw_i)$ and $\omega(v_iv_j) \leq \omega(w_jv_j)$ for each $v_iv_j \in E$. Then*

$$\mathrm{r}_{R'}\left(\frac{R'}{I((\Sigma H)_\omega)}\right) = \sharp\,\{\textit{minimal weighted vertex covers of } H_\epsilon\}.$$

For example, an example of a Cohen-Macaulay weighted suspension $(\Sigma P_2)_\omega$ of

$$(P_2)_\epsilon = (\ v_1 \overset{2}{\text{———}} v_2 \overset{3}{\text{———}} v_3\ )$$

is

$$\begin{array}{ccccc} w_1 & & w_2 & & w_3 \\ \vert^{5} & & \vert^{3} & & \vert^{4} \\ v_1 & \overset{2}{\text{———}} & v_2 & \overset{3}{\text{———}} & v_3. \end{array}$$

Then since we calculated the minimal weighted vertex covers of $(P_2)_\epsilon$ above, we have

$$\mathrm{r}_{R'}\big(R'/I((\Sigma P_2)_\omega)\big) = \sharp\,\{\text{minimal weighted vertex covers of } (P_2)_\epsilon\} = 2.$$

4 *Future Work*

I will continue to study the Cohen-Macaulayness of monomial ideals I by studying $\Upsilon(I)$ combinatorially and I will also investigate when I is sequentially Cohen-Macaulay and describe the minimal free resolution of I using these ideas.

Furthermore, I am interested in classifying certain other monomial ideal constructions (e.g., path ideals [3, 5]) for edge-weighted graphs which are Cohen-Macaulay, and then computing the type combinatorially.

References

[1] M. F. Atiyah and I. G. Macdonald. *Introduction to commutative algebra*. Addison-Wesley Publishing Co., Reading, Mass.-London-Don Mills, Ont., 1969.

[2] Winfried Bruns and Jürgen Herzog. *Cohen-Macaulay rings revised edition*, volume 39 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 1993.

[3] Daniel Campos, Ryan Gunderson, Susan Morey, Chelsey Paulsen, and Thomas Polstra. Depths and Cohen-Macaulay properties of path ideals. *J. Pure Appl. Algebra*, 218(8):1537–1543, 2014.

[4] Reinhard Diestel. *Graph theory*, volume 173 of *Graduate Texts in Mathematics*. Springer, Berlin, fifth edition, 2018. Paperback edition of [MR3644391].

[5] Bethany Kubik and Keri Sather-Wagstaff. Path ideals of weighted graphs. *J. Pure Appl. Algebra*, 219(9):3889–3912, 2015.

[6] Ezra Miller. Alexander duality for monomial ideals and their resolutions. *arXiv preprint math/9812095*, 1998.

[7] W. Frank Moore, Mark Rogers, and Keri Sather-Wagstaff. *Monomial ideals and their decompositions*. Universitext. Springer, Cham, 2018.

[8] Chelsey Paulsen and Keri Sather-Wagstaff. Edge ideals of weighted graphs. *J. Algebra Appl.*, 12(5):1250223, 24, 2013.

[9] Keri Sather-Wagstaff. Homological algebra notes. *Unpublished lecture notes. Available at* `http://ssather.people.clemson.edu/notes.html`, 2009.

[10] Rafael H. Villarreal. Cohen-Macaulay graphs. *Manuscripta Math.*, 66(3):277–293, 1990.